PRE-ALGEBRA (8TH) · UNIT 2: EQUATIONS AND EXPRESSIONS

# Phone Plan Showdown

*Write and solve equations to find the best cell phone plan for any customer.*

**CP1**

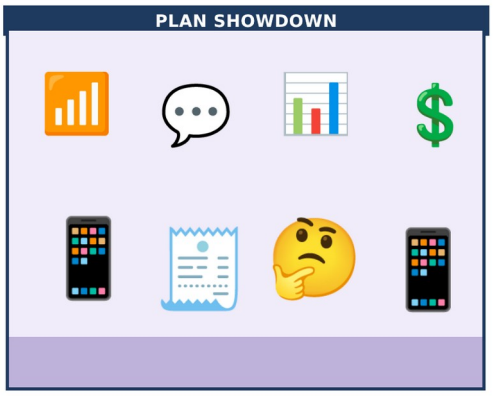
2–3 class periods

Name: _____ Date: _____ Period: _____

## The Scenario

A phone store has hired you to help customers choose a plan. Each plan has a monthly fee and a price per gigabyte of data, and customers want to know which plan is cheapest for them.

You will write expressions and solve equations to compare the plans and build a guide that any customer can use.




**Plan A**
$15 per month
+ $5 per GB

**Plan B**
$30 per month
+ $2 per GB

**Plan C**
$60 per month
unlimited data

## Learning Goals

- I can write and simplify expressions using the distributive property and like terms.
- I can solve multi-step equations with variables on both sides.
- I can tell when an equation has one, no, or infinitely many solutions.

## Project Timeline

| When | What you will do |
|---|---|
| Day 1 | Plans as Expressions |
| Day 2 | Solving Equations |
| Day 3 | Special Cases & Recommendations |
| Final | Plan Comparison Guide + reflection |

## Part 1: Plans as Expressions

**Day 1 ·** Let g be the GB used each month.

**1.** Write expressions for the monthly cost of Plans A, B, and C.

**2.** A family has 3 lines on Plan A, each using g GB, plus a $10 family fee. Write and simplify the total cost.

**3.** A customer uses g GB in month 1 and 3 more GB in month 2. Write and simplify the two-month cost on Plan A.

## Part 2: Solving Equations

**Day 2 ·** Show each step.

**4.** Solve 15 + 5g = 30 + 2g and explain what the solution means.

**5.** Solve 3(g − 2) + 4 = 2g + 7.

**6.** The family's Plan A bill was $160. How many GB did each line use?

## Part 3: Special Cases & Recommendations

**Day 3 ·** Compare all three plans.

**7.** How many solutions does each equation have? Explain. (a) 2(5g + 15) = 10g + 30 (b) 5g + 15 = 5g + 20

**8.** For which amounts of data is Plan B the cheapest of all three plans?

**9.** A customer averages 12 GB each month. Which plan do you recommend? Show the costs.

## Final Product: Plan Comparison Guide

**Your final product must include:**

- Simplified cost expressions, including the family plan
- Solutions to every equation with steps
- Ranges of data where each plan is cheapest
- A customer guide (table or graph) with recommendations
- An example of a no-solution and an infinite-solution equation from the plans

## Reflection

**R1.** What does it mean in context when an equation has no solution?

**R2.** How did combining like terms help you compare plans?

## Scoring Rubric

| Criteria | 4 – Exceeds | 3 – Meets | 2 – Approaching | 1 – Beginning |
|---|---|---|---|---|
| **Mathematical Accuracy** | All calculations and solutions are correct; answers include units and are checked. | Nearly all work is correct; minor errors do not change conclusions. | Several errors; some conclusions are affected. | Many errors or missing work. |
| **Reasoning & Justification** | Clear, complete explanations connect every answer to the situation; uses precise vocabulary. | Explanations are mostly clear and connected to the situation. | Explanations are brief or partly unclear. | Little or no explanation. |
| **Application & Design** | Expressions and equations model the plans correctly, solutions are correct, and solution types are explained. Goes beyond the minimum with a creative, realistic choice. | Expressions and equations model the plans correctly, solutions are correct, and solution types are explained. | Design is incomplete or only partly uses the math from the unit. | Design is missing or unrelated to the unit math. |
| **Communication & Presentation** | Customer guide clearly states which plan is best for each amount of data and why. Polished, organized, and visually engaging. | Customer guide clearly states which plan is best for each amount of data and why. Organized and easy to follow. | Product is missing required parts or is hard to follow. | Product is mostly missing. |